# FINAL EXPLANATION: BIOLOGY GRADE 10

*Student Assessment Document*

| FINAL EXPLANATION: BIOLOGY GRADE 10 — SUB-STRAND 1.1: CELL STRUCTURE \| FINAL EXPLANATION ASSESSMENT | |
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| **Student Assessment Document** | |
| **Student Name** | ________________________________________ |
| **Class** | ________________________________________ |
| **Date** | ________________________________________ |

## INSTRUCTIONS FOR STUDENTS

WHAT IS A FINAL EXPLANATION?

A Final Explanation is your opportunity to answer the driving question using everything you have learned across all 12 lessons — backed by evidence from investigations, microscopy, readings, and discussions. It is not a summary of notes. It is a scientific argument: you make a claim, support it with evidence, and explain your reasoning.

DRIVING QUESTION: "Why do cells in our body look so different from each other — and how does the shape of a cell determine the job it can do?"

REMEMBER THE PHENOMENON: A Form 1 student in Nairobi cuts their finger while slicing onions. Seconds later, bleeding slows. A clot forms. Days later, new pink skin replaces the scab. This was not one process — it was many, each carried out by a completely different type of cell. Your Final Explanation must return to this phenomenon and fully explain how and why it happens at the cellular level.

HOW TO USE THIS DOCUMENT:

- Read each section prompt carefully before you write.
- Study the exemplar response — notice how it uses specific evidence, names real structures, and always connects back to the phenomenon.
- Your own response should match or exceed the depth shown in the exemplar.
- Use the rubric at the end to check your work before submitting.

WRITING TIPS:

- Write in full sentences and organised paragraphs.
- Name specific organelles, cell types, and their functions — do not be vague.
- Use connective phrases such as "This explains why…", "The evidence shows that…", "Because of this structure…", and "Returning to the phenomenon…"
- Aim for depth over length: one well-reasoned paragraph is worth more than three vague ones.

## Section 1 — The Foundation: Cells Are the Basic Unit of Life

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| **Begin your explanation by establishing what cells are and why understanding them is essential to explaining the phenomenon. Answer: What is the cell theory, and why does it matter for understanding how the cut finger heals? Who are some key scientists whose work helped us understand cells, and what did they discover?** | The starting point for explaining the healing cut is the cell theory, which states three fundamental ideas: all living things are made of cells, the cell is the basic unit of life, and all cells come from pre-existing cells. This theory was built over centuries of scientific work. Robert Hooke first observed and named cells in 1665 when he examined cork under a primitive microscope. Later, Matthias Schleiden and Theodor Schwann extended this to plants and animals, and Rudolf Virchow established that cells only arise from other cells — a principle that is essential to wound healing.<br><br>Returning to the phenomenon: when the student in Nairobi cuts her finger while slicing onions, the damage is happening at the cellular level — skin cells are torn open, blood cells are released, and tissue is disrupted. Healing is only possible because the body can produce new cells to replace the damaged ones. Without the principle that cells come from pre-existing cells, there would be no way to explain how new pink skin appears days later. Every event in the healing process — clot formation, infection control, tissue rebuilding — is carried out by cells. This is why the cell theory is not just a definition to memorise, but the very foundation of the explanation. |

| Section 2 — Inside the Cell: Organelles as a Coordinated System | |
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| **Explain the internal structure of a cell by describing the key organelles and their functions. Then connect each organelle to the healing process. Answer: What structures did you observe through microscopy, and how do they work together as a system to allow healing to occur? Your explanation must mention at least five organelles by name.** | Through microscopy investigations, we observed that cells are not simple, empty blobs — they are highly organised systems containing specialised structures called organelles, each with a specific role. The cell membrane forms the outer boundary of every animal cell, controlling which substances enter and exit; during healing, it allows chemical signals — called cytokines — to pass in, alerting the cell that damage has occurred nearby. The nucleus acts as the control centre, containing DNA that carries the instructions for producing every protein the cell needs; when healing begins, the nucleus activates specific genes that direct the cell to divide or to manufacture repair proteins. The mitochondria generate ATP, the energy currency of the cell; without a constant energy supply, no healing activity could be sustained. Ribosomes read the instructions from the nucleus and assemble proteins — including the collagen that fibroblasts deposit to rebuild skin tissue after a wound. Finally, the Golgi apparatus packages and dispatches these proteins to where they are needed, functioning like a postal sorting office inside the cell.<br><br>These organelles do not work in isolation — they form an integrated system. When a fibroblast near the wound in Nairobi needs to produce collagen, the nucleus issues the instruction, ribosomes build the protein, the endoplasmic reticulum transports it, the Golgi packages it, and the cell membrane releases it into the wound site. This coordinated sequence explains why wound repair is both rapid and precise. |

| Section 3 — Animal Cells vs. Plant Cells: Why Flexibility Matters for Healing | |
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| **Compare the structure of animal cells and plant cells, identifying the key structural differences. Then explain why the fact that human skin cells are animal cells — not plant cells — matters for how wounds heal. Consider what would happen if our skin cells had the same structures as the onion cells the student was** | During our comparative microscopy investigation, we examined onion epidermal cells alongside cheek cells and observed clear structural differences between plant and animal cells. Plant cells, like those in the onion the student was slicing, have a rigid cell wall made of cellulose surrounding the cell membrane, a large central vacuole that fills most of the cell's interior, and — in green plant cells — chloroplasts for photosynthesis. Animal cells have none of these structures. Instead, they are bounded only by the flexible cell membrane, contain smaller vacuoles, and rely on food consumed from outside for energy.<br><br>These differences are critical to the healing process. Because human skin cells — epithelial cells — have no rigid cell wall, they are flexible and can change shape, migrate across a wound surface, and squeeze between other cells to close a gap. After the cut on the student's finger, it is precisely this flexibility that allows new epithelial cells to migrate from the wound edges and cover the exposed tissue. If our skin cells had rigid cell walls like the onion cells being sliced, this migration would be impossible — cells would be locked in fixed positions and unable to close the wound. The absence of a cell wall in animal cells is therefore not a |

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| **cutting when she got hurt.** | weakness but a critical adaptation that makes tissue repair possible. Additionally, the lack of chloroplasts means animal cells must produce energy through cellular respiration in the mitochondria — an efficient system that can rapidly scale up energy production during the high-energy demands of healing. |

| Section 4 — Cell Specialisation: How Shape Determines Job | |
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| **This is the heart of your explanation. Describe at least FOUR different specialised cell types involved in healing the cut finger. For each cell: (a) describe its shape and key structural features, (b) explain how that shape directly enables its function, and (c) explain its specific role in healing the wound. Use the phrase 'structure determines function' and show what it means in practice.** | A central principle of biology is that structure determines function — the physical shape and internal organisation of a cell directly determines what job it can perform. Nowhere is this clearer than in the coordinated cellular response to a wound.<br><br>Platelets are tiny, irregular cell fragments with no nucleus. Because they have no nucleus, they take up very little space, allowing billions of them to circulate freely in the bloodstream. Their jagged, sticky surface allows them to clump together rapidly at the wound site and form a plug. Without the elongated, nucleated structure of a full cell slowing them down, platelets can aggregate within seconds — explaining why the student's bleeding slowed almost immediately after the cut.<br><br>Red blood cells (erythrocytes) are biconcave discs — curved inward on both sides — and, uniquely among human cells, they have no nucleus in their mature form. Losing the nucleus frees up space for haemoglobin, the protein that carries oxygen. The biconcave shape maximises surface area for gas exchange and allows red blood cells to bend and squeeze through the narrowest capillaries to deliver oxygen to the oxygen-hungry healing tissue.<br><br>Neutrophils and macrophages are white blood cells that can change shape dramatically. Macrophages extend long projections called pseudopodia to engulf and destroy bacteria that try to enter through the open wound. If these cells had a fixed, rigid shape, they could not surround and engulf pathogens — their shape-shifting ability is the structural basis of their function as the body's first line of cellular defence.<br><br>Fibroblasts are elongated, spindle-shaped cells rich in rough endoplasmic reticulum and ribosomes. This internal structure reflects their function: fibroblasts are protein-manufacturing machines. Their elongated shape and protein-production machinery allow them to secrete large quantities of collagen — the structural protein that forms the scaffold of new skin tissue. The pink, slightly raised scar that eventually replaces the scab is largely collagen laid down by fibroblasts.<br><br>Epidermal cells (keratinocytes) are flat and tightly interlocking, like tiles on a floor. This flattened shape allows them to pack closely together, forming a continuous, waterproof barrier — the skin. After the wound, new keratinocytes migrate from the edges, driven by chemical signals, and layer over the repaired tissue to restore this barrier. Their flat, adherent shape is perfectly suited to forming a sealed surface.<br><br>In every case, you can see that the shape of the cell is not accidental — it is the physical expression of the cell's function. |

| Section 5 — From Cells to Healing: Levels of Organisation and the Full Explanation | |
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| **Zoom out from individual cells to explain how cells organise into tissues, organs, and organ** | Individual specialised cells do not act alone — they organise into increasingly complex levels of biological organisation, and it is this organisation that makes a response as sophisticated as wound healing possible. Cells of the same type group together to form tissues: keratinocytes form epithelial tissue, fibroblasts and collagen fibres form connective tissue, and neurons form nervous |

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| **systems to carry out the complex process of healing. Then write a concluding paragraph that fully answers the driving question using all the evidence from this explanation. Your conclusion must explicitly reference the phenomenon and leave no part of the driving question unanswered.** | tissue. These different tissues combine to form the skin — an organ. The skin itself is part of the integumentary system, which works alongside the circulatory system (delivering platelets, red blood cells, and white blood cells) and the immune system (coordinating the infection-fighting response). Healing a cut is therefore not the work of one cell, one tissue, or even one organ — it is a whole-organism response orchestrated across multiple levels of organisation simultaneously.<br><br>Returning now to the driving question: why do cells in our body look so different from each other, and how does the shape of a cell determine the job it can do?<br><br>The answer, supported by all the evidence gathered across this unit, is this: all cells in the human body contain the same DNA, but they activate different genes, which causes them to develop different structures — different shapes, different organelles, different surface proteins. This process, called cell specialisation, produces cells that are exquisitely fitted to specific functions. A red blood cell's disc shape and lack of nucleus maximise oxygen-carrying capacity. A macrophage's flexible, shape-shifting membrane enables it to engulf pathogens. A fibroblast's ribosome-rich interior equips it to manufacture collagen. In every case, structure is not separate from function — structure IS function.<br><br>When the student in Nairobi cut her finger while slicing onions, her body did not send one type of cell to fix the problem. It sent many — each with a different shape, each with a different job, and each performing that job because of its shape. Platelets plugged the wound, white blood cells fought infection, fibroblasts rebuilt the tissue, and keratinocytes sealed the surface. Each of these cells looks completely different under a microscope, yet all arose from the same organism, carrying the same DNA. The diversity of cell shape is not a paradox — it is the body's greatest strength. Specialisation creates efficiency, and efficiency makes healing possible. |

| **RUBRIC** | | | |
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| **Criterion** | **Excellent (4)** | **Proficient (3)** | **Developing (1–2)** |
| Cell Theory & Scientific Foundation | Accurately states all three components of cell theory and correctly attributes key discoveries to relevant scientists (e.g., Hooke, Schleiden, Schwann, Virchow). Clearly explains why cell theory is foundational to the phenomenon, with explicit connection to new cell production during healing. | States at least two components of cell theory and names at least one scientist with a correct discovery. Makes a general connection between cell theory and the healing phenomenon. | Mentions cells as the basic unit of life but does not state cell theory accurately or fully. Reference to scientists is missing or incorrect. Connection to the phenomenon is weak or absent. |
| Organelle Structure & Coordinated Function | Names and accurately describes the function of five or more organelles. Clearly explains how multiple organelles work together as an integrated system during wound healing. Uses specific language (e.g., 'the nucleus activates healing genes,' 'ribosomes synthesise collagen proteins'). | Names and describes the function of three to four organelles. Makes some connection between organelle function and healing, but the explanation of how organelles work as a system is incomplete. | Names fewer than three organelles or describes their functions vaguely (e.g., 'the nucleus controls the cell'). Little or no connection is made between organelle function and the healing process. |
| Plant vs. Animal Cell Comparison | Accurately identifies at least three | Identifies two structural differences | Identifies one or fewer structural |

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| | structural differences between plant and animal cells. Provides a thorough, logically reasoned explanation of why the absence of a cell wall in animal cells enables cell migration and wound closure. Uses the onion-cutting context to anchor the comparison. | between plant and animal cells. Explains in general terms why animal cell flexibility is important for healing, but the reasoning is not fully developed. | differences, or differences stated are inaccurate. The connection between cell structure and healing ability is missing or unclear. |
| Cell Specialisation: Structure Determines Function | Describes four or more specialised cell types involved in healing. For each, accurately links a specific structural feature (shape, organelle composition, or surface property) to a specific function, and explains the cell's precise role in healing. Uses the phrase 'structure determines function' meaningfully. | Describes two to three specialised cell types and makes a reasonable link between structure and function for most of them. The role of each cell in healing is mentioned but may lack specificity. | Describes fewer than two cell types, or the link between structure and function is absent or incorrect. The explanation reads as a list of cell types without connecting shape to job. |
| Levels of Organisation & Driving Question Answer | Accurately describes at least four levels of biological organisation (cell → tissue → organ → organ system) with correct examples from the healing context. The concluding paragraph explicitly and fully answers both parts of the driving question using evidence from all sections. The phenomenon is directly referenced in the conclusion. | Describes two to three levels of organisation with generally correct examples. The conclusion addresses the driving question but may answer only one of its two parts, or the connection to the phenomenon is brief. | Levels of organisation are missing or only one level is mentioned. The conclusion does not clearly answer the driving question, or the phenomenon is not referenced. |